

REVIEW

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The safety and efficacy of GD2-targeted CAR-T cells in patients with neuroblastoma: a systematic review and meta-analysis

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Abstract

Objectives Neuroblastoma (NB) is a common and life-threatening pediatric solid tumor with a poor prognosis, especially in high-risk patients. Although anti-disialoganglioside GD2 (GD2) monoclonal antibodies improve survival, relapse and resistance remain major challenges. This study aimed to evaluate the safety and efficacy of GD2-targeted chimeric antigen receptor (CAR)-T-cell therapy in NB patients.

Methods A literature search was conducted in PubMed, Embase, Scopus, and Web of Science until September 30, 2025. Eligible studies were clinical trials that evaluated the efficacy and safety of GD2-targeted CAR-T-cell therapy in NB patients. Two reviewers independently performed study selection, data extraction, and risk-of-bias assessment. The random-effects meta-analysis model was used to calculate pooled amounts.

Results Eight studies that included 146 patients with NB were included. The pooled complete response (CR) rate was 39.57% (21.17–57.96), and the partial response (PR) rate was 15.83% (5.02–30.45). Additionally, the rates of progressive disease (PD) and stable disease (SD) were 20.9% (3.06–46.67) and 30.76% (12.81–51.91), respectively. The most common “any grade” adverse events (AEs) was anemia, at 97.43% (81.51–100), and the most common “grade ≥ 3” AEs was neutropenia, at 93.46% (72.65–100). Subgroup analyses revealed that CAR-T-cell generation and its components influenced efficacy and safety.

Conclusion CAR-T-cell therapy targeting the GD2 antigen is promising for treating NB. However, its efficacy is moderate, and treatment can lead to hematologic toxicities such as anemia, neutropenia, and thrombocytopenia, which require careful monitoring.

Keywords Neuroblastoma, Chimeric antigen receptor, GD2, Immunotherapy, CAR-T-cell therapy

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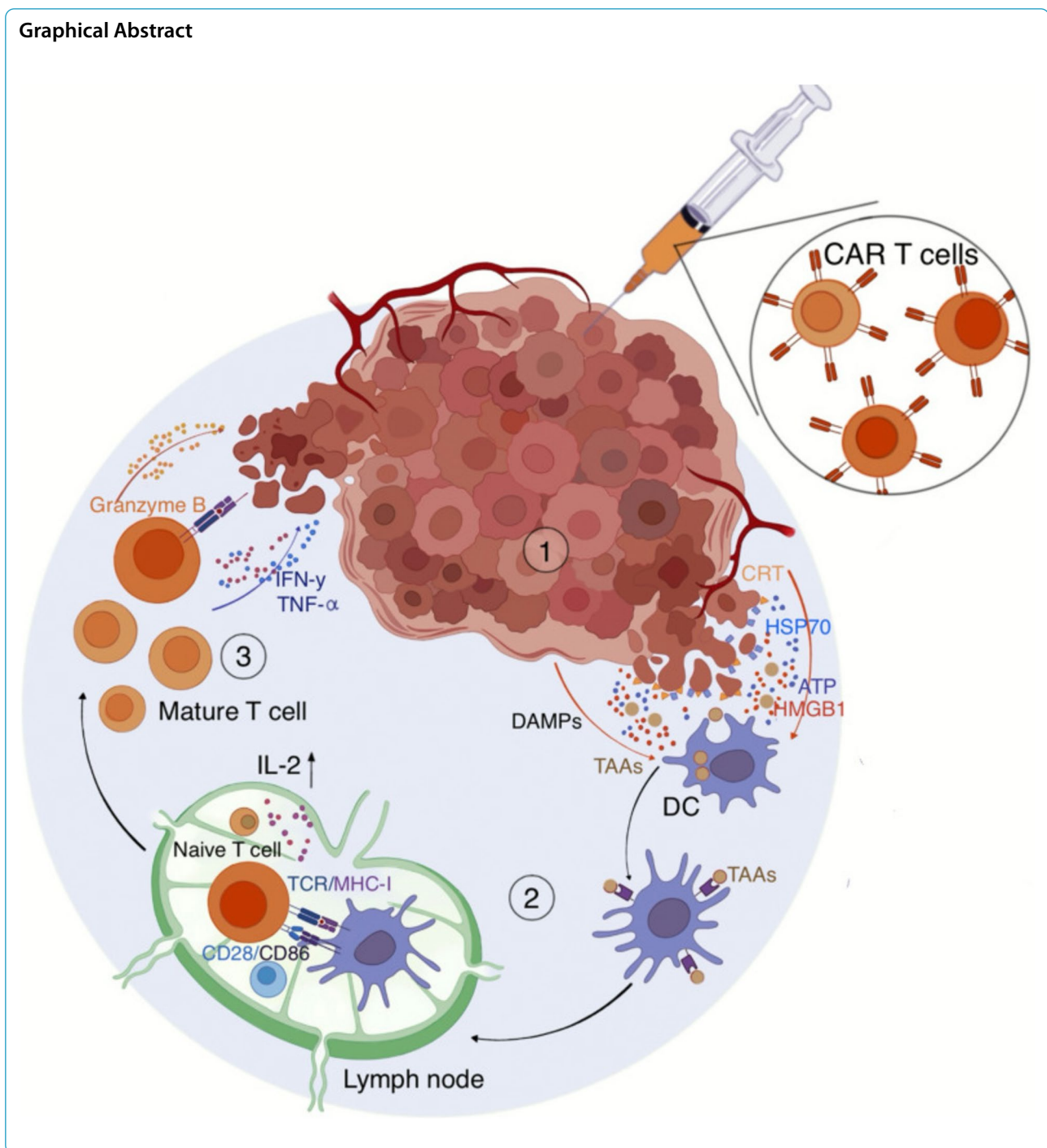
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Graphical Abstract



Introduction

NB, a solid tumor originating from the neuroendocrine tissue of the sympathetic nervous system, is a common cause of pediatric cancer mortality [1, 2]. Typically, identified within the first decade of life, this condition is the most prevalent solid tumor outside of the brain in children [3]. It is frequently detected in the perinatal stage,

affecting 8% of patients under fifteen years of age. In the US, there are over 600 cases of this pediatric neoplasm annually [4–6].

NB presents with significant heterogeneity, ranging from asymptomatic to extensive tumors causing systemic symptoms. Patient outcomes vary, with some patients experiencing spontaneous regression and others

facing unfavorable progression, metastasis, or death [6]. Adrenal tumors are common in newborns and infants, whereas toddlers and older children often have metastatic disease requiring multimodal intensive therapy, such as surgery, chemotherapy, radiotherapy, autologous stem cell transplant, differentiation therapy, and monoclonal antibody-based immunotherapy [7]. NB accounts for 11% of all cancer-related deaths. Approximately 50% of patients are diagnosed with high-risk conditions, and their 5-year event-free survival (EFS) is between 40 and 50% [3, 8]. Patients who survive sometimes experience long-term consequences of intensive therapy, including hearing loss and growth retardation [9]. Approximately 50% of patients have metastases at diagnosis, primarily affecting the bone marrow (BM), bone, and regional lymph nodes [10]. This population needs innovative medicines to increase survival and reduce morbidity.

The disialoganglioside GD2 is highly expressed in NB cells, and targeting it with monoclonal antibodies has been linked to a significantly increased survival rate in high-risk patients. These findings highlight the importance of target antigens and the sensitivity of NBs to immunotherapy [11, 12]. A phase III clinical trial has assessed the use of anti-GD2 monoclonal antibodies in the treatment of high-risk NB patients. This therapy has shown promising results, leading to other immunotherapeutic approaches. Patients receiving adjuvant anti-GD2 monoclonal antibodies with IL-2, GM-CSF, and retinoic acid presented a 2-year EFS increase of 46–66% compared with that of patients receiving retinoic acid alone [11]. These findings establish a new paradigm for NB treatment involving immunotherapy. Despite improved survival rates since anti-GD2 antibody adoption, approximately 50% of patients will relapse and die from their disease [11]. In addition, 20% of patients are refractory to induction therapy and may never receive anti-GD2 antibodies [13, 14].

CAR cells are genetically engineered T cells that express synthetic receptors against specific tumor antigens to identify and target cancer cells [15, 16]. They have four major components: an extracellular or antigen recognition domain, a hinge, a transmembrane domain, and an intracellular signaling domain. CAR function independently of the major histocompatibility complex, allowing for a more extensive range of therapies [17]. The US Food and Drug Administration has approved six CAR-T-cell therapies for hematological cancers, including B-cell lymphoblastic leukemia and B-cell non-Hodgkin lymphomas [18]. CAR-T cells can traverse the blood–brain barrier, whereas antibodies often cannot enter the central nervous system [19–21].

Following the discovery that T cells also target GD2, a disialoganglioside that is extensively expressed in most

NBs, cellular immunotherapies, such as genetically engineered T lymphocytes that express anti-GD2 CAR, have been developed and are now under investigation. Relapsed NB patients have shown safety and antitumor efficacy with anti-GD2 CAR-T cells due to their combination of antigen specificity and cytolytic activity [22, 23]. Early-phase studies have shown that CAR-T cells have potential in treating NB, with multiple objective responses in clinical trials [24–27]. CAR-T-cell efficacy in NB is limited by factors such as the tumor microenvironment (TME), cell exhaustion, and T-cell persistence and potency, which may lead to therapeutic resistance, unlike its success in hematological malignancies [24, 28, 29]. The design of CAR-T cells for NB presents challenges such as inadequate T-cell persistence, a scarcity of tumor-specific targets, and an immunosuppressive TME [24, 30, 31].

The field of CAR-T-cell engineering is rapidly progressing to improve the effectiveness of tumor targeting [32–34]. NB is a suitable platform for evaluating these new treatments, as immunotherapy has proven successful for this condition. This systematic review provides an overview of the clinical experience with CAR-based NB therapies, discussing improvements in CAR in various areas to address clinical issues and suggesting strategies to enhance the potency of CAR immunotherapy.

Method

Objective

This systematic review and meta-analysis sought to investigate the safety and efficacy of CAR-T-cell therapy for patients with NB. The study followed the suggested parameters specified in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. Moreover, the paper is currently undergoing a systematic review and has been officially registered with the International Prospective Register of Systematic Reviews (PROSPERO, CRD42024542364) and assigned a unique registration code.

Eligibility criteria

In this study, we established specific inclusion and exclusion criteria, based on the PICO framework, to ensure the selection of clinically relevant studies. These criteria defined eligible patient populations, interventions (GD2-targeted CAR-T therapy), outcomes (treatment response and safety), and study designs, as follows:

Participants

This study focused on human patients diagnosed with NB. No restrictions were applied on the basis of age or disease stage.

Interventions

The studies included in this review focused on autologous CAR-T-cell therapies designed to target the GD2 antigen in NB patients. Although there are differences in CAR designs, dosing strategies, and lympho-depletion protocols across studies, all of these methods share a common feature: GD2 is the central therapeutic target.

Outcomes

The primary outcomes of interest were treatment response, including CR, PR, SD, and PD. Treatment responses (CR, PR, SD, and PD) were extracted as reported in each study and were generally based on the International Neuroblastoma Response Criteria (INRC), which define standard clinical and radiologic benchmarks for assessing disease status in NB. Where studies did not explicitly reference INRC, response definitions were consistent with its framework. In addition, we evaluated safety outcomes by examining the frequency and severity of AEs related to the therapy, such as cytokine release syndrome (CRS), neutropenia, anemia, and thrombocytopenia.

The criteria for including studies in this review were as follows:

- 1- Published in English
- 2- This study was conducted on human NB patients, irrespective of disease stage (new, refractory, or relapsed).
- 3- Patients who received autologous CAR-T-cell therapy specifically targeting GD2
- 4- At least one clinical outcome of interest, including treatment response (CR, PR, SD, PD) or AEs, was reported.
- 5- The study designs included clinical trials (phases I–II), prospective or retrospective case series, and conference abstracts reporting original data.

The exclusion criteria for this study were as follows:

1. Non-English publications
2. Studies conducted in animals or in vitro
3. Studies enrolling patients with malignancies other than NB
4. Studies not involving NB patients or not assessing CAR-T-cell therapy
5. Reviews, editorials, letters to the editor, book chapters, and duplicate publications
6. Studies lacking outcome data relevant to efficacy or safety.

Search strategy

We conducted a literature search on PubMed, Embase, Scopus, and the Web of Science until September 30, 2025. The search was limited to studies published in English that involved human participants. To maximize the sensitivity of the search, we combined MeSH terms with free-text keywords related to NB and CAR-T-cell therapy. The search terms included “NB,” “CAR T cell,” “CAR-T,” “CAR,” “GD2,” and “GD2-targeted immunotherapy.” Boolean operators such as AND and OR were applied to structure the queries effectively, and the search syntax was used to fit each database’s indexing format. We did not exclude any studies on the basis of design during the initial screening. The complete search strategy for each database is available in Supplementary Appendix 1.

Study selection process

All the retrieved records were imported into EndNote V.20, and duplicate entries were removed. Two independent reviewers, P. Lorestani and M. Ahmadpour, selected the study in two stages. First, they screened titles and abstracts to identify potentially relevant articles. In the second stage, full-text articles were reviewed to assess their eligibility on the basis of the predefined inclusion and exclusion criteria. Any disagreements between the two reviewers were resolved through discussion, and when consensus could not be reached, a third reviewer, M. Dashti, was consulted to make the final decision.

Data extraction and variables

Two reviewers, P. Lorestani and F. Maleki, independently extracted data from the eligible studies via a standardized template. The extracted variables included bibliographic details (first author, publication year), study characteristics (trial phase, sample size, follow-up duration, conditioning chemotherapy, number of prior treatment lines), and patient demographics (median age, sex distribution, disease status: newly diagnosed, refractory, or relapsed). The details of the CAR-T-cell therapy, including the CAR construct and generation, cell source, dosage, target antigen, and site of disease involvement, were also recorded. Reported outcomes were categorized as either clinical responses—such as CR, PR, SD, and PD—or safety outcomes, including CRS, neutropenia, anemia, thrombocytopenia, and other AEs. Any disagreements in data extraction were resolved by consultation with a third reviewer, M. Dashti, who reviewed the sources independently and made the final determination.

Quality assessment

The methodological quality of the included studies was assessed via the Joanna Briggs Institute (JBI) Critical

Appraisal Checklist, with the specific version selected according to each study design [35]. Two independent reviewers, P. Lorestani and M. Ahmadpour, evaluated the risk of bias on the basis of domains such as participant selection, outcome measurement, and completeness of follow-up. Each criterion was rated as “yes,” “no,” “unclear,” or “not applicable,” and studies were not excluded on the basis of their risk of bias. Disagreements between the reviewers were resolved through discussion, and M. Dashti provided the final judgment when needed.

Meta-analysis

We used Stata version 17 to carry out the statistical analysis. Given the expected differences between studies, a random-effects model (DerSimonian–Laird method) was used to generate more generalizable results. To assess heterogeneity across studies, we used Cochran's Q test and the I^2 statistic; I^2 values above 50% were considered to reflect moderate to substantial heterogeneity. The pooled estimates are reported with their corresponding 95% confidence intervals. In addition, subgroup analyses were conducted to better understand the sources of variation in safety outcomes, with a focus on relevant biological and therapeutic factors.

Results

A total of 1045 records were initially identified through database searches. After 385 duplicates were removed via EndNote, 660 unique articles remained. Two independent reviewers screened the titles and abstracts, excluding 617 records that did not meet the inclusion criteria. The remaining 43 studies underwent full-text evaluation; however, two could not be retrieved despite repeated attempts. Of the 41 accessible full texts, 33 were excluded for nonrelevant outcomes, ineligible study designs, or insufficient data. Ultimately, eight studies met all the criteria and were included in the qualitative synthesis and the meta-analysis [25, 30, 36–41]. The PRISMA flow diagram illustrates the detailed selection process (Fig. 1).

Baseline characteristics

Eight eligible clinical trials involving 146 NB patients who received autologous CAR-T cells at different doses of 0.13×10^6 cells/kg and 1×10^7 cells/kg on the basis of patient weight and 1×10^7 cells/m² and 1×10^9 cells/m² on the basis of surface area were identified. All patients were administered autologous CAR-T cells, and the median follow-up range was between one and five years. Trials were published between 2011 and 2025, encompassing phases such as I and II. As detailed in Table 1, three studies took place in the USA, two in China, one in the UK, and two in Italy. The sex of the 114 participants was delineated in six studies, with 68 (59.65%) male

participants and 46 (40.35%) female participants, and the median age of the 95 enrolled patients ranged from 1.7 to 28 years. Baseline disease burden and minimal residual disease (MRD) levels were variably reported across trials. Where specified, most participants had measurable or bulky disease at enrollment, although some trials included patients treated in minimal residual disease settings after consolidation therapy. Table 1 depicts the features and characteristics of the included studies. Furthermore, the JBI scale was employed to assess the risk of bias in the included studies (Supplementary materials). Table 2 summarizes the clinical trials for CAR-T cells in NB.

Safety

Our examination of all the evaluable AEs among the 146 patients revealed that the most common “any grade” AEs manifested in the following order: anemia, neutropenia, thrombocytopenia, and CRS (Fig. 2. A–D). Severe (Grade ≥ 3) AEs were predominantly hematologic, with neutropenia being the most frequent, followed by thrombocytopenia, anemia, and CRS (Fig. 3; Table 2). Egger's regression asymmetry test suggested no considerable publication bias concerning any grade or grade ≥ 3 AEs ($p > 0.05$). Funnel plots were used to visually assess potential publication bias in both safety and efficacy analyses. Each point represents an individual study, with the effect size plotted on the x axis and the standard error on the y axis. In the absence of publication bias, studies are expected to be symmetrically distributed around the pooled estimate, forming an inverted funnel shape. In our analysis, the plots for both any-grade and grade ≥ 3 AEs as well as for treatment response outcomes appeared generally symmetrical, suggesting no major evidence of small-study effects or publication bias. This visual impression was further supported by Egger's regression test ($p > 0.05$ for all endpoints). Minor asymmetry observed in a few plots likely reflects the limited number of included trials ($n=8$) and differences in CAR-T constructs or dosing regimens rather than systematic bias (Figures S1–S4).

According to the subgroup analysis, a significant difference was observed in any grade CRS rate among the subgroups of CAR-T-cell generation (lowest: 0.33 for the third generation, highest: 0.90 for the fourth generation, $p=0.00$) and OX40 (also known as CD134 or TNFRSF4) expression (lowest: 0.12 for expression, highest: 0.56 for the lack of OX40, $p=0.03$). In addition, subgroup analysis revealed a significant alteration in the grade 1–2 CRS rate among the subgroups of CAR-T-cell generation (lowest: 0.42 for the third generation, highest: 0.90 for the fourth generation, $p=0.00$), OX40 expression (lowest: 0.12 for expression,

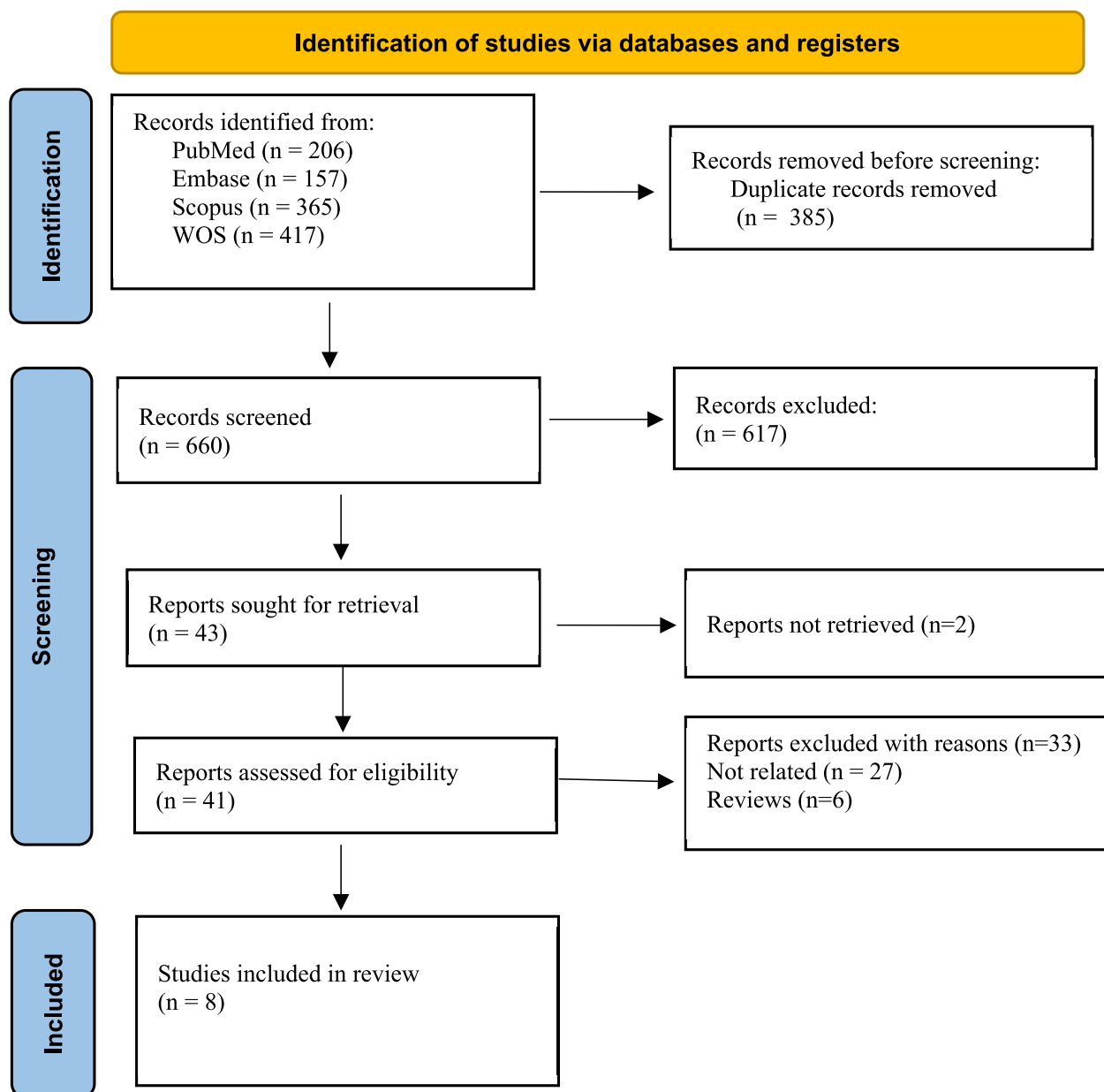


Fig. 1 PRISMA flow diagram summarizing the systematic search and study selection process

highest: 0.69 for the lack of OX40, $p=0.00$), and inducible caspase 9 (iCasp9) incorporation (lowest: 0.19 for the absence of iCasp9, highest: 0.77 for the presence, $p=0.00$).

For any grade neutropenia rate, we found a significant difference among the subgroups of CAR-T-cell generation (lowest: 0.25 for the second generation, highest: 0.90 for the fourth generation, $p=0.00$) and iCasp9 incorporation (lowest: 0.39 for the absence of iCasp9, highest: 0.99 for the second generation, $p=0.00$) (Table 3).

Efficacy

This study revealed that 39.57% (21.17–57.96) of patients achieved a CR to treatment and that 15.83% (5.02–30.45) of patients achieved a PR (Fig. 4A–D). Since the heterogeneity was significant regarding CR, PR, and SD, we considered a random effects model to determine the pooled effect sizes. The sensitivity analysis revealed no significant variations in the pooled estimates ($p>0.05$), except for CR, and SD (Figures S3, S4).

Subgroup analysis revealed a substantial alteration in the SD rate among the subgroups of CAR-T-cell

Table 1 Characteristics of the included studies

Author, Year	Region	Clinical trial ID	Phase of trial	Number of patients	Median age (range)	Sex (number of males)	Treatment strategy-dose escalation (CAR per kg)
Louis, 2011 [25]	USA	NCT00085930	I	19	7 (3–20)	9 (47%)	Dose level 1 (2×10^7 cells/m ²) Dose level 2 (5×10^7 cells/m ²) Dose level 3 (1×10^8 cells/m ²)
Heczey, 2017 [30]	USA	NCT01822652	I	11	6.5 (4.1–23.6)	3 (27%)	Cohort 1: 10×10^6 , 100×10^6 Cohort 2: 100×10^6 , 150×10^6 Cohort 3: 150×10^6
Straathof, 2020 [39]	UK	NCT02761915	I	12	8 (2–15)	7 (58.33%)	1×10^7 /m ² 1×10^8 /m ² 1×10^9 /m ²
Liu, 2022 [37]	China	NCT02992210	I/II	19	N/A	N/A	2.93×10^6 cells/kg
Ramakrishn, 2022 [38]	USA	NCT02107963	I	13	(8–28)	N/A	Four dose levels, up to 1×10^7 cells/kg
Yu, 2022 [40]	China	NCT02765243	I	10	4.5	7 (70%)	$0.13\text{—}34 \times 10^6$ /kg
Del Bufalo, 2023 [36]	Italy	NCT03373097	I/II	27	6.7	18 (67%)	3×10^6 cells/kg 6×10^6 cells/kg 10×10^6 cells/kg
Locatelli F, 2025 [41]	Italy	NCT03373097	I/II	35	6.3 (1.7–18.6)	24 (69%)	3×10^6 cells/kg 6×10^6 cells/kg 10×10^6 cells/kg

generation (lowest: 0.05 for first-generation, highest: 0.40 for third-generation, $p=0.01$), OX40 expression (lowest: 0.17 for the lack of OX40, highest: 0.63 for expression, $p=0.01$), and iCasp9 incorporation (lowest: 0.18 for the presence, highest: 0.63 for the absence of iCasp9, $p=0.01$). In terms of CR, a significant difference was observed among the subgroups in which CAR-T cells were generated (lowest: 0.16 for the first generation, highest: 0.63 for the fourth generation, $p=0.00$) (Table 3).

Discussion

Our study shed light on the safety and efficacy of CAR-T-cell therapy in patients with NB, encompassing eight clinical trials and 146 patients. The findings highlight both the promise and challenges of this innovative treatment approach, demonstrating that while CAR-T-cell therapy has shown potential in treating NB, its efficacy remains limited.

Efficacy of CAR-T-cell therapy

Our findings revealed that one-third of patients responded positively to CAR-T-cell therapy, underscoring unsatisfactory results in attenuating regressing malignancies. This response heterogeneity may partly reflect differences in baseline tumor burden. Previous evidence suggests that patients with lower levels of MRD at infusion exhibit improved CAR-T-cell expansion and persistence, leading to higher response rates and reduced relapse risk [42, 43]. The inconsistent reporting of MRD

status across included trials represents an additional source of variability in efficacy outcomes. Some studies reported that fewer than half of all patients experienced SD after CAR-T-cell infusion, and few patients achieved CR during long-term follow-up [30, 40]. On the other hand, in 2011, Louis et al. reported no evidence of disease (NED) as the best response in almost half of the patients [25]. However, this rate of NED patients reached approximately 25% in the long-term follow-up [25]. Moreover, challenges remain in achieving more robust and durable CR, such as poor T-cell persistence, a lack of truly tumor-specific targets, and an immunosuppressive TME [44]. In other words, the immunosuppressive TME, characterized by regulatory T cells, myeloid-derived suppressor cells, and inhibitory cytokines, further limits the efficacy of CAR-T-cell therapy in solid tumors [24, 44]. Innovative strategies, such as combining CAR-T-cell therapy with checkpoint inhibitors or modifying CAR-T cells to resist suppressive signals, are currently being explored to overcome these barriers [45].

Del Bufalo et al. reported that among patients who received the recommended dose of 10×10^6 CAR-T cells/kg, the 3-year overall survival rate was 60%, and the 3-year EFS rate was 36% [36]. Additionally, they discovered that patients with a low disease burden had significantly longer progression-free survival than did those with a high disease burden. At 3 years, the EFS rate was 58% for patients with a low disease burden versus 0% for those with a high disease burden [36].

Table 2 Summary of clinical trials for CAR-T cells in NB

Author, Year	Trial Status	Type of malignancy (new, refractory/relapsed)	Relapsed sites	Site of disease involvement	Target	Construct	Conditioning Chemotherapy
Louis, 2011 [25]	Active, Not Recruiting	Eight had no evidence of active disease Eleven had active relapsed/refractory disease	Bone, BM	N/A	GD2	Activated T cells (CAR-ATCs) Epstein-Barr virus-specific T cells (CAR-CTLs)	N/A
Heczey, 2017 [30]	Active, Not Recruiting	Relapsed/refractory	Bone, BM, liver, and lymph nodes	One patient had an unresectable stage 3 pelvic tumor Ten patients had stage 4 (metastatic) disease	GD2	iC9.GD2-CAR3 T cells incorporating CD28 and OX40 costimulatory endodomains	Cohort 1: None Cohorts 2 & 3: Cyclophosphamide and fludarabine (Cy/Flu)
Straathof, 2020 [39]	Completed	Relapsed/refractory	Bone, BM, soft tissue	Bone, bone marrow, soft tissue	GD2	GD2 antibody K666 linked via IgG1 Fc spacer to a CD28 endodomain	Cyclophosphamide alone or Cy/Flu
Liu, 2022 [37]	Unknown Status	Relapsed/refractory	Bone, BM, and lymph nodes	N/A	GD2	safety-engineered lentiviral CAR (45CAR) carrying the following intracellular signaling domains: CD28-CD27-CD3ζ-iCasp9	Cy/Flu
Ramakrishna, 2022 [38]	Completed	N/A	N/A	N/A	GD2	GD2-CAR.OX40.28.z.ICD9	Cyclophosphamide-based lympho-depletion Cy/Flu
Yu, 2022 [40]	Completed	Refractory (8/10) or relapsed (2/10)	Bone, BM, soft tissue masses, lymph nodes	Primary tumor sites (spinal canal, abdomen), BM, bones, lymph nodes	GD2	CD28/4-1BB/CD3ζ-iCasp9	
Del Bufalo, 2023 [36]	Recruiting	Relapsed or refractory	N/A	Bone (21 patients), BM (12 patients), lymph nodes (11 patients), abdomen (4 patients), paravertebral area (7 patients), thorax/pleura (2 patients), liver (1 patient)	GD2	iCasp9.2A.GD2.CD28-4-1BB.zeta	Cy/Flu lympho-depleting chemotherapy
Locatelli F, 2025 [41]	Completed	Refractory or Relapsed		Bone (29 patients) BM (18 patients) Lymph nodes (15 patients) Abdomen (15 patients) Paravertebral/thoracic region (8 patients) Pleura/lung (4 patients) Liver (2 patients)	GD2	GD2-targeting CAR containing CD28-4-1BB-CD3ζ costimulatory domains and an inducible caspase-9 (iC9) suicide gene	Dinutuximab β/naxitamab in combination with chemotherapy

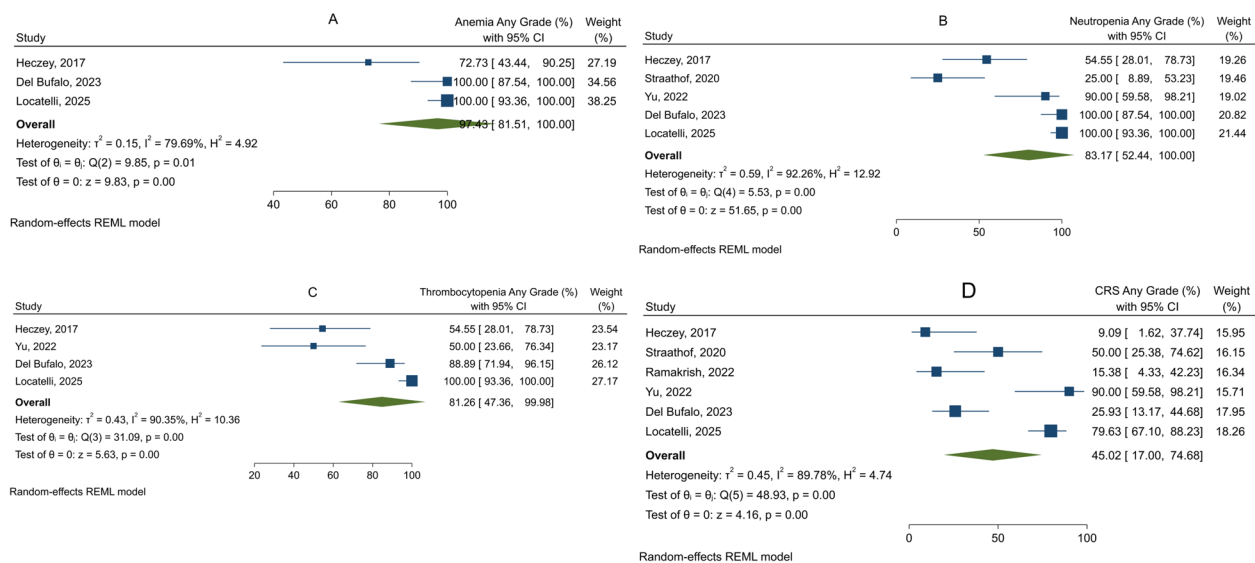


Fig. 2 Pooled rates of any-grade AEs associated with GD2-targeted CAR-T-cell therapy, including anemia, neutropenia, thrombocytopenia, and CRS. The forest plots demonstrate that hematologic toxicities were the most frequent AEs across studies, reflecting the expected on-target/off-tumor marrow suppression due to lympho-depleting regimens: **A** Anemia (any grade); **B** neutropenia (any grade); **C** thrombocytopenia (any grade), and **D** CRS (any grade)

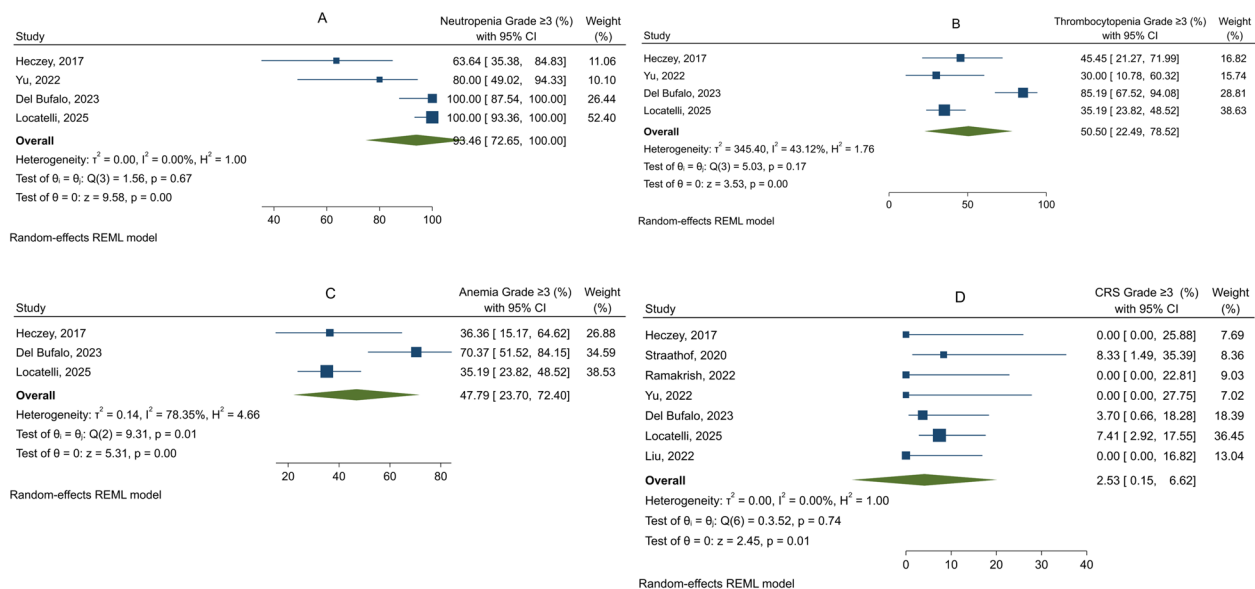


Fig. 3 Pooled proportions of severe AEs following CAR-T-cell therapy at all doses in patients with NB: **A** Neutropenia (Grade ≥ 3); **B** Thrombocytopenia (Grade ≥ 3); **C** Anemia (Grade ≥ 3), and **D** CRS (Grade ≥ 3)

Safety of CAR-T-cell therapy

Our results demonstrated that anemia was the most common AEs of any grade and that neutropenia was the most prevalent AEs of Grade ≥ 3. Some studies have shown that CAR-T-cell therapy targeting the GD2 antigen is both feasible and safe for treating patients with relapsed or refractory high-risk NB [44, 46, 47]. Powderly

et al. reported that an anti-B7-H3 monoclonal antibody (MGA271) has a beneficial safety profile in adult patients with malignancies [46]. Similarly, in another clinical trial performed by Heczey et al. in 2017, they reported critical expansion of CD11b/CD33/CD45/CD163 in myeloid cells in all patients. As a result, they concluded that CAR-T cells are a safe choice [30]. On the other hand,

Table 3 Subgroup analysis based on different variables

Outcome	Variable	Subgroup (number of studies)	Proportion	p
Any grade CRS	CART cells generation	First-generation (n=0)	-	0.00
		Second-generation (n=1)	0.50	
		Third-generation (n=4)	0.33	
		Fourth-generation (n=1)	0.90	
	OX40 expression	Yes (n=2)	0.12	0.03
		No (n=3)	0.56	
Grade 1–2 CRS	iCasp9 incorporation	Yes (n=3)	0.65	0.16
		No (n=3)	0.22	
	CART cells generation	First-generation (n=0)	–	0.00
		Second-generation (n=1)	0.42	
		Third-generation (n=4)	0.42	
		Fourth-generation (n=1)	0.90	
Any grade neutropenia	OX40 expression	Yes (n=2)	0.12	0.00
		No (n=3)	0.69	
	iCasp9 incorporation	Yes (n=3)	0.77	
		No (n=3)	0.19	
Grade ≥ 3 neutropenia	CART cells generation	First-generation (n=0)	–	0.00
		Second-generation (n=1)	0.25	
		Third-generation (n=3)	0.87	
		Fourth-generation (n=1)	0.90	
	OX40 expression	Yes (n=1)	0.55	0.51
		No (n=3)	0.73	
Grade ≥ 3 neutropenia	iCasp9 incorporation	Yes (n=3)	0.99	0.00
		No (n=2)	0.39	
	CART cells generation	First-generation (n=0)	–	0.86
		Second-generation (n=0)	–	
		Third-generation (n=3)	0.91	
		Fourth-generation (n=1)	0.80	
PD	OX40 expression	Yes (n=1)	0.64	0.08
		No (n=3)	0.96	
	iCasp9 incorporation	Yes (n=2)	0.94	
		No (n=1)	0.64	
PD	CART cells generation	First-generation (n=1)	0.21	0.43
		Second-generation (n=0)	–	
		Third-generation (n=3)	0.26	
		Fourth-generation (n=0)	–	
	OX40 expression	Yes (n=2)	0.37	0.37
		No (n=1)	0.21	
SD	iCasp9 incorporation	Yes (n=2)	0.12	0.37
		No (n=2)	0.37	
	CART cells generation	First-generation (n=1)	0.05	0.01
		Second-generation (n=0)	–	
		Third-generation (n=4)	0.40	
		Fourth-generation (n=1)	0.40	
SD	OX40 expression	Yes (n=2)	0.63	0.01
		No (n=3)	0.17	
	iCasp9 incorporation	Yes (n=4)	0.18	
		No (n=2)	0.63	

Table 3 (continued)

Outcome	Variable	Subgroup (number of studies)	Proportion	p
CR	CAR T cells generation	First-generation (n = 1)	0.16	0.00
		Second-generation (n = 0)	–	
		Third-generation (n = 3)	0.35	
		Fourth-generation (n = 1)	0.63	
	OX40 expression	Yes (n = 1)	0.18	0.31
		No (n = 3)	0.36	
	iCasp9 incorporation	Yes (n = 4)	0.40	0.17
		No (n = 1)	0.18	

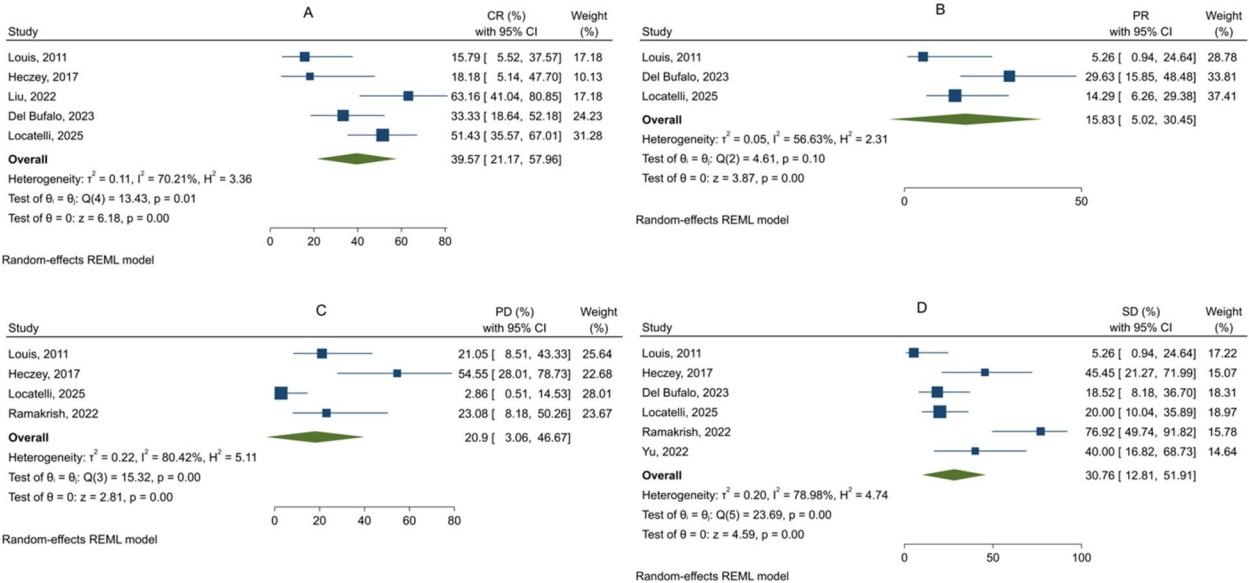


Fig. 4 Treatment response outcomes, including CR, PR, PD, and SD rates, following CAR-T-cell therapy at all doses for patients with NB

some studies have reported that the most common acute toxicities associated with CAR-T-cell therapy in NB include CRS and neurotoxicity [36, 40]. CRS occurs as a systemic inflammatory response triggered by rapid and robust activation of CAR-T cells upon encountering their target antigen. The binding of CAR-T cells to GD2-expressing tumor cells induces a surge of pro-inflammatory cytokines, such as IL-6, IFN- γ , and TNF- α , released from both T cells and bystander immune cells. This cytokine storm leads to the characteristic symptoms of fever, hypotension, and organ dysfunction. Improvements in later-generation CAR-T constructs have helped to mitigate this risk [48]. Specifically, the incorporation of dual costimulatory domains (CD28 and 4-1BB) allows for more controlled activation and better persistence, while reducing excessive cytokine secretion. Moreover, suicide or safety switches such as iCasp9 enable

clinicians to rapidly eliminate CAR-T cells in severe CRS cases, markedly improving safety profiles in recent trials. Advances in manufacturing, resulting in more defined T-cell subsets and optimized dosing, have also contributed to reducing CRS severity compared with earlier generations [49]. For example, Yu et al. reported CRS in Grades 1 and 2 and neutropenia and fever in Grade ≥ 3 as the most common AEs [40]. In addition, some studies have reported that hematologic AEs, including anemia, are the most common safety issue, followed by neurologic AEs [30, 36]. These are class-specific AEs observed with CAR-T-cell therapies across cancer types [48, 50, 51]. The most prominent long-term toxicities observed with CAR-T-cell therapy include cytopenias (low blood cell counts) and hypogammaglobulinemia (low antibody levels) [50]. Notably, management strategies for CRS typically involve the use of tocilizumab, an IL-6 receptor

antagonist, and corticosteroids to mitigate severe inflammatory responses [48, 52]. Ensuring prompt identification and treatment of CRS and neurotoxicity is crucial for improving patient outcomes and safety [51]. Furthermore, persistent B-cell depletion and immunoglobulin deficiency have been reported in a significant proportion of NB patients even years after infusion of CAR-T cells [53–56]. As a result, careful monitoring and management of these acute toxicities are needed, as they can be severe in some cases [44].

Comparison with current treatments

The current treatment modalities for high-risk NB, including chemotherapy, surgery, radiotherapy, and immunotherapy, have improved survival rates but are often associated with significant long-term morbidities [11, 57]. Although the current treatments such as chemotherapy and anti-GD2 monoclonal antibodies have significantly improved outcomes for some patients, they also result in substantial long-term morbidities, such as hearing loss and growth retardation, which impact quality of life [57, 58]. The potential of CAR-T-cell therapy to reduce these long-term toxicities, if optimized, offers a significant advantage over conventional therapies [59]. Some studies have shown that anti-GD2 monoclonal antibody therapy combined with cytokines and retinoic acid results in a 2-year EFS increase from 46 to 66% [60, 61]. However, approximately 50% of patients still relapse or are refractory to induction therapy, necessitating alternative treatment strategies [57, 58, 62].

In addition, unlike hematological malignancies, where CAR-T-cell therapy has demonstrated impressive efficacy, the treatment of solid tumors poses unique challenges due to factors such as the TME, T-cell exhaustion, and persistence [63, 64]. Solid tumors pose unique challenges for CAR-T cells, including antigen heterogeneity, physical barriers to infiltration, and an immunosuppressive metabolic milieu [63, 65].

Our study indicates that while CAR-T-cell therapy offers a novel approach, its efficacy in NB remains limited compared with the successes observed in hematologic cancers [59, 66, 67]. The modest response rates and significant toxicities suggest that further optimization of CAR-T-cell design, targeting, and delivery is crucial to enhance its therapeutic potential in NB [24, 45, 52].

Limitations and future directions

This study has several limitations. The small number of included trials and patients may limit the generalizability of our findings. The heterogeneity in trial design, patient populations, CAR-T-cell constructs, and dosages further complicates direct comparisons and pooled analyses. In addition, the follow-up duration varied across studies,

potentially affecting the assessment of long-term outcomes and late-onset AEs.

Future research should focus on improving CAR-T-cell persistence and function in the TME, identifying and validating new tumor-specific targets, and developing combination therapies to increase efficacy and reduce resistance. Advances in CAR-T-cell engineering, such as the incorporation of additional costimulatory domains, bispecific CAR, and advanced CAR, may address current limitations and improve clinical outcomes.

Conclusion

In conclusion, our meta-analysis highlights the potential of CAR-T-cell therapy for NB, although its current efficacy is limited and associated with significant toxicity. Our findings highlight both the therapeutic promise and current limitations of GD2-directed CAR-T-cell therapy in NB. Rigorous multicenter phase III trials are needed to validate safety and long-term efficacy, paving the way for integration of CAR-T strategies into the multimodal treatment of high-risk pediatric solid tumors.

Abbreviations

AEs	Adverse events
BM	Bone marrow
CAR	Chimeric antigen receptor
CR	Complete response
CRS	Cytokine release syndrome
Cy/Flu	Cyclophosphamide and fludarabine
EFS	Event-free survival
GD2	Disialoganglioside GD2
iCasp9	Inducible caspase 9
INRC	International Neuroblastoma Response Criteria
JBI	Joanna Briggs Institute
NB	Neuroblastoma
NED	No evidence of disease
PD	Progressive disease
PR	Partial response
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROSPERO	International Prospective Register of Systematic Reviews
SD	Stable disease
TME	Tumor microenvironment

Supplementary Information

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Additional file 1.

Additional file 2.

Author contributions

SA, wrote the manuscript. MAH, AZ, BRJ, AK, and MD contributed to the methodology and formal analysis. NN, PL, ZK, and MA reviewed and revised the final manuscript. FM, and SA are correspondence(s). All the authors read and approved the final manuscript.

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Availability of data and materials

The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study is deemed exempt from receiving ethical approval.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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